PRESENTER'S GUIDE

Stock Explorer

A practice stock market that explains itself. This guide explains **every part of the dashboard twice**: once in simple words with an everyday picture, and once in proper finance words.

Who this is for

You are going to present Stock Explorer to visitors. You do **not** need to know anything about the stock market yet. This guide starts at the very beginning (what a share is) and builds up, one idea at a time. Each page uses words explained on the pages before it, so **read it in order**.

Read it once through. Then keep it with you while you present, and use the contents list to jump to a page.

How every topic is laid out

IN SIMPLE WORDS an everyday picture, for a friend who has never heard of the topic.

IN FINANCE WORDS the proper explanation, with the correct terms, as a finance professional would say it.

ON THE SCREEN exactly where you will see it in the app, and what it looks like.

Contents

The story in one minute and the golden rules	2
Part 1. The basics: companies, shares, exchanges	3
Reading a price: volume, candles, trends, risk	4
Your portfolio, practice trading, ETFs and bonds	6
Futures and options	7
Part 2. The screen, part by part	9
The chart and its tools	10
Part 3. The six indicators	12
Part 4. The five strategies and the ML model	14
The back-test, and why honesty matters	17
Using the site, and a 3-minute demo	19
Questions you will be asked, and a glossary	20

The path through this guide

STEP 1
The basics
Shares, prices, volume, candles, trends, risk, portfolios. Pages 3 to 7.

STEP 2
What you can trade
ETFs, bonds, futures and options. Pages 6 to 8.

STEP 3
The screen
A map of every box, and the chart tools. Pages 9 to 11.

STEP 4
The readers
Six indicators, five strategies, the ML model and the back-test. Pages 12 to 18.

Important. Stock Explorer is an **educational project**. All money in it is **virtual** and nothing on it is investment advice. The prices come from Yahoo Finance and can be a few minutes late. Say this at the start of every presentation.

START HERE

The story in one minute

Learn these lines first. Everything else in the guide helps you answer questions about them.

Say it like this:
"This is Stock Explorer, a practice stock market. Real companies, real prices, but **pretend money**, so you can learn without losing anything. On the left is your practice portfolio. In the middle is the price chart of any company you search for. On the right are six **indicators**: small gauges that read the chart for you. Along the bottom are **five strategies** from a professional exam called the CMT, and each one gives its reading, a vote. The dial on the left combines the five votes. And in the corner there is a chat helper that explains any word you do not know."

The three golden rules

1 It is practice money

Nothing here is real. You cannot win or lose actual rupees. Say "practice money" or "virtual money", never "invest your money".

2 It describes, it never advises

The site shows what the charts are saying. It never says "buy this". If a visitor asks "should I buy?", the answer is: "That is not what it is for. It shows what the signals read; the decision is yours, and a real decision needs a real adviser."

3 Be honest about the limits

In our own tests the strategies were right about **50 times out of 100**, which is the same as flipping a coin. We show that openly on purpose. It makes the site trustworthy.

Say this, not that

Instead of	Say
"The price **will** fall."	"The reading **leans down**." (a lean is not a promise)
"You should buy this."	"The strategy says Up. Whether to act on it is up to you."
"Invest Rs 10,000."	"Practise with Rs 10,000 of virtual money."
"The ML model predicts tomorrow."	"The ML model gives a modelled chance. In tests it was about as good as a coin toss."
"This stock is a winner."	"This company passes the trend test the strategy uses."

How this guide is ordered

Part 1, the basics (pages 3 to 8). Companies, shares, prices, volume, candles, trends, risk, portfolios, ETFs, bonds, futures and options. You cannot understand an indicator without these.

Part 2, the screen (pages 9 to 11). A map of every box on the dashboard, and the tools on the chart.

Part 3, the indicators (pages 12 to 14). The six gauges on the right.

Part 4, the strategies (pages 14 to 18). The five strategies, the ML model, the dial and the back-test.

The end (pages 19 to 20). A demo script, likely questions and a glossary.

PART 1. THE BASICS

Companies, shares and the stock market

Start with these five ideas. Every later page leans on them.

1 A company, and a share (also called a stock)

IN SIMPLE WORDS

Imagine a pizza shop. The shop is cut into a million tiny slices, and anyone can own a slice. A **share** is one slice of a company. If you own a slice, you own a tiny piece of the shop, and if the shop does well, your slice becomes more valuable.

IN FINANCE WORDS

A **share** (or **stock**) is a unit of **ownership** in a company. Companies sell shares to raise money to grow. A shareholder owns part of the company and may get a share of the profit, called a **dividend**. A company whose shares anyone can buy is called **listed**.

ON THE SCREEN

You search for a company by name (a small spelling slip is fine) and its price chart appears. You can buy and sell shares of it with practice money.

2 Why does a share price go up and down?

IN SIMPLE WORDS

Think of a popular toy at school. If everyone wants it and few have it, the price goes up. If nobody wants it, the price falls. A share price is simply what buyers and sellers agree on right now.

IN FINANCE WORDS

Price is set by **supply and demand** in the market. More buyers than sellers pushes the price up. More sellers than buyers pushes it down. News, company profits, interest rates and the general mood of investors (**sentiment**) all change how many people want to buy or sell.

ON THE SCREEN

The big number at the top of the chart is the latest price. Next to it is the change since yesterday's close, in rupees and in percent (green for up, red for down).

3 The stock exchange: NSE and BSE

IN SIMPLE WORDS

An exchange is like a big, well-organised market (a mela with strict rules) where all buying and selling of shares happens in one place, fairly and openly.

IN FINANCE WORDS

A **stock exchange** is a regulated marketplace where listed shares are traded. India has two main ones: the **NSE** (National Stock Exchange) and the **BSE** (Bombay Stock Exchange). Many big companies are listed on both. The regulator that watches over them is **SEBI**.

ON THE SCREEN

You can search both listings. A company's short code, called a **ticker symbol** (like RELIANCE or TCS), is shown next to its name. It is the company's nickname on the exchange.

4 An index: Nifty 50 and Sensex

IN SIMPLE WORDS

An index is a **class scoreboard**. Instead of looking at 50 students one by one, you look at their average marks. The Nifty 50 is the average of the 50 biggest companies on the NSE, so it tells you how 'the market' is doing overall.

IN FINANCE WORDS

An **index** tracks a basket of shares. The **Nifty 50** covers 50 large NSE companies, weighted by size. The **Sensex** covers 30 large BSE companies. They are used as the **benchmark**: to say whether a company did better or worse than the market.

ON THE SCREEN

The 'Relative strength' strategy compares a company with the Nifty 50 (or Sensex for a BSE listing).

5 Market hours, and the 'last close'

IN SIMPLE WORDS

The market is like a shop with opening hours. Outside them nothing is bought or sold, so the price stays where it stopped.

IN FINANCE WORDS

The Indian market trades **9:15 am to 3:30 pm, Monday to Friday**, except holidays. The **last close** is the final price of the previous session. Outside hours the screen shows it, with a note that the market is closed.

ON THE SCREEN

The line under the price says 'Market closed (opens Monday at 9:15 am), last close' or tells you the market is open. While it is open, parts of the screen refresh themselves every few seconds. Prices come from Yahoo Finance and can be a few minutes late.

PART 1. THE BASICS

Reading a price: volume, candles, trends and risk

Now the language of the chart. Every indicator later is built from these.

1 Volume

IN SIMPLE WORDS

Volume is how many shoppers walked into the mall today. A big crowd means lots of interest. A price move that many people joined is more believable than one a few people made.

IN FINANCE WORDS

Volume is the number of shares traded in a period. High volume on a move **confirms** it. Low volume suggests weak conviction. Volume is shown separately from price because it measures participation, not value.

ON THE SCREEN

The bars under the chart. Green bars are days the price rose, red bars are days it fell.

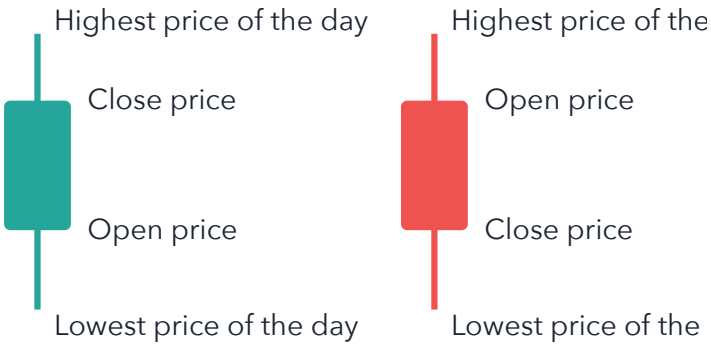
2 Open, High, Low, Close and the candle

IN SIMPLE WORDS

Each day has a little story: where the price **started**, the **highest** and **lowest** it reached, and where it **finished**. A candle draws that story as one small shape. Green means the day ended higher than it began. Red means lower.

IN FINANCE WORDS

The four daily prices are **OHLC**: open, high, low, close. A **candlestick** shows them: the thick **body** runs from open to close and the thin **wicks** reach the high and low. A green body means close above open.



Left: a green candle (the price rose). Right: a red candle (the price fell).

ON THE SCREEN

Every bar on the main chart is one candle: a day, a week or a month, depending on the Day, Week and Month buttons.

3 Trend: up, down or sideways

IN SIMPLE WORDS

A trend is the direction a price has been walking. Up like climbing stairs, down like descending them, or sideways like walking along a corridor.

IN FINANCE WORDS

A **trend** is the general direction of price over time: an **uptrend** (rising highs and lows), a **downtrend**, or a **sideways** (range-bound) market. Trend-following aims to go with the trend, not against it.

ON THE SCREEN

Most strategies and indicators are really different ways of asking: is there a trend, and how strong is it?

4 Risk and return

IN SIMPLE WORDS

Return is what you might gain. Risk is how bumpy the road is on the way, and how much you could lose. A very bumpy road can lead somewhere great, or nowhere. Bigger possible gains come with bigger possible losses.

IN FINANCE WORDS

Return is the percentage gain or loss on money invested. **Risk** is the uncertainty of that return, often measured by **volatility** (how much the price swings). Higher expected return generally needs accepting higher risk. There is no free lunch.

5 Diversification

IN SIMPLE WORDS

Do not carry all your eggs in one basket. If you drop it, they all break. Putting eggs in different baskets means one accident does not ruin everything.

IN FINANCE WORDS

Diversification means spreading money across different companies, sectors and asset types, so one bad result hurts less. It lowers the risk specific to one company, but it does not remove the risk of the whole market falling.

ON THE SCREEN

The ring on the left shows how your practice money is spread across stocks, ETFs, bonds, futures, options and cash.

PART 1. THE BASICS

Your portfolio, practice trading, ETFs and bonds

What you hold, how the site lets you practise, and two things to hold besides shares.

1 Portfolio, cash, profit and return

IN SIMPLE WORDS

Your **portfolio** is your bag of things you own, plus the cash you have not spent. If you start with Rs 1,00,000 and your bag is worth Rs 1,05,000 a week later, you are up Rs 5,000. That is a **5% return**.

IN FINANCE WORDS

A **portfolio** is the collection of all your holdings plus **cash**. Its value is cash plus the current market value of everything held. **Profit or loss (P&L)** is value minus what you put in, and the **return** is that as a percentage. If you buy the same share at different prices, your **average buy price** is the total paid divided by the units.

ON THE SCREEN

'Portfolio value' at the top shows the total and the percent change since you started. 'Bought: what you hold' lists each holding with its profit or loss. 'Sold: recent sales' lists your last six sales.

2 Paper trading

IN SIMPLE WORDS

Cricketers practise in the **nets** before a real match. Paper trading is the nets for the stock market: real prices, pretend money, nothing to lose, plenty to learn.

IN FINANCE WORDS

Paper trading (also called virtual or simulated trading) means placing trades with pretend money at real market prices, to learn how buying, selling, profit and loss work without financial risk. It is a standard way to practise a strategy before committing real capital.

ON THE SCREEN

You pick your starting practice money (Rs 50,000, 1 lakh, 5 lakh, 10 lakh or your own amount). The Buy, Sell and Exit all buttons below the chart place practice trades. No brokerage or tax is charged here; real trading has both.

3 ETF (exchange-traded fund)

IN SIMPLE WORDS

An ETF is a **lunchbox with a bit of everything**. Instead of buying one food item, you buy one box that has many items inside. One ETF can hold a whole basket of companies, or gold, or silver.

IN FINANCE WORDS

An **ETF** is a fund that holds a basket of assets (shares of an index, gold, silver or bonds) and is bought and sold on the exchange like a share. A **Nifty 50 ETF** simply copies the Nifty 50. ETFs give instant **diversification** with one purchase and usually low costs.

ON THE SCREEN

In the Buy pop-up, the 'ETFs and bonds' tab lists Nifty 50, Bank, IT, Gold and Silver ETFs, among others.

4 Bond

IN SIMPLE WORDS

A bond is **lending money**. You lend to the government or a company, and they promise to give it back on a set date, plus a little extra (interest) for using it. It is like lending your friend Rs 100 and getting back Rs 106 next year.

IN FINANCE WORDS

A **bond** is a loan to an issuer (a government or company) that pays regular **interest** (the **coupon**) and returns the **principal** at **maturity**. Bond prices move opposite to interest rates. Government bonds are generally lower-risk than company bonds, and bonds are usually calmer than shares.

ON THE SCREEN

The same 'ETFs and bonds' tab includes bond ETFs such as the Bharat Bond ETF and government-security (G-Sec) ETFs, plus 'liquid' ETFs that behave like cash.

Cannot sell what you do not own. In the practice account you can only sell shares you hold, and you can only buy what your cash allows. If cash runs short, the Buy button says so.

PART 1. THE BASICS

Futures and options

These are called **derivatives**: their value comes from (is *derived* from) something else, such as a share. They are riskier, so go slowly.

Future

IN SIMPLE WORDS

A farmer and a shopkeeper agree **today** that the shopkeeper will buy 100 kg of mangoes next month at Rs 80 a kilo, **whatever the price turns out to be**. Both are **bound** to the deal. If the price rises to Rs 100, the shopkeeper gained. If it falls to Rs 60, the farmer gained.

IN FINANCE WORDS

A **futures contract** is an agreement to buy or sell an asset at a fixed price on a fixed date. It is an **obligation** for both sides. You put down only a small deposit, the **margin**, so gains and losses are large compared with what you put in: this is **leverage**.

Long and short

IN SIMPLE WORDS

Long = you gain if the price goes **up**. **Short** = you gain if it goes **down**, like betting a ticket price will fall.

IN FINANCE WORDS

Going **long** profits from a rise; going **short** profits from a fall. In this app you can go long or short in futures. Shares can only be bought, then sold.

Margin, lot and expiry

IN SIMPLE WORDS

Margin is a deposit, like the one you pay to rent a bike. A **lot** is a fixed pack size, like eggs sold by the dozen. **Expiry** is the date the deal ends, like a milk packet's date.

IN FINANCE WORDS

Margin: a deposit that backs the position (here 15% of the contract's value). A **lot**: the fixed quantity in one contract. **Expiry**: the contract's end date (here the last Tuesday of the month). If losses use up the margin, the position is closed automatically.

Option: call and put

IN SIMPLE WORDS

You pay a small fee to **reserve** a phone at today's price for a week. If the price jumps, you use your reservation and save money. If it drops, you just walk away and lose only the small fee. An option is a **right**, not a must.

A **call** = the right to **buy** (you hope the price goes up).
A **put** = the right to **sell** (you hope it goes down).

IN FINANCE WORDS

An **option** gives the buyer the **right but not the obligation** to buy (a **call**) or sell (a **put**) an asset at a fixed price, the **strike price**, on or before **expiry**. The price paid for that right is the **premium**. The most the buyer can lose is the premium, but it can lose all of its value quickly, especially near expiry.

	Future	Option (bought)
What you hold	An obligation	A right
Up-front cost	Margin (a deposit)	Premium (a fee, gone if unused)
Worst case	Large losses compared with the margin	Lose the whole premium
Direction	Long or short	Call (up) or put (down)

How the app does it. Real exchange prices for futures and options are not freely available, so the app **calculates** them from the live share price with standard formulas (futures price grows at a 6.5% safe rate until expiry; options use the Black-Scholes formula). Lot sizes and margin are **simplified for learning** (a lot is worth about Rs 2 lakh and margin is 15%). In this app, options can only be **bought**, not sold. Say that these are **estimates, not exchange quotes**.

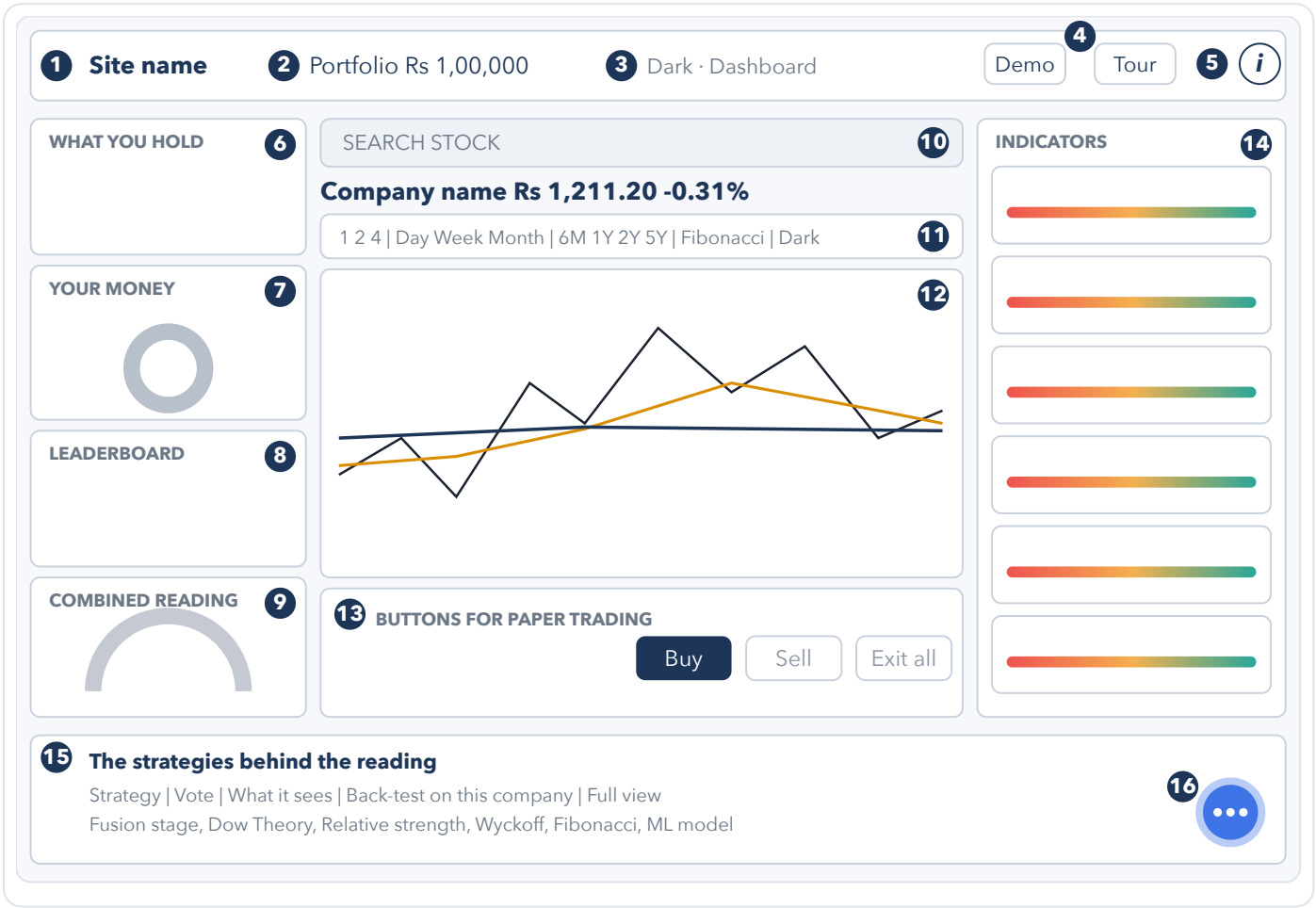
ON THE SCREEN

Click **Buy** and the pop-up has four tabs: Stocks, ETFs and bonds, Futures, Options. Choose what to trade, how many, and press the button. After an order the pop-up **stays open**, so you can place the next one, and the page behind updates when you close it.

PART 2. THE SCREEN

The dashboard, part by part

One screen holds everything. Match the numbers on the map to the list below.



#	What it is	What it does, in plain words
1	Site name and crest	The school crest and the name of the site.
2	Portfolio value	Everything you own, in rupees, with the percent gained or lost since you started. Grey 0.00% means no change yet.
3	Dark mode and Dashboard view	Dark mode changes the colours. Dashboard view switches between this wide screen and a simpler phone-style layout.
4	Demo and Tour	**Demo** plays a short demo by itself: it searches a company, changes the chart, opens a strategy and makes a practice buy, with a big caption and a spoken voice, and you can change its speed like a video. Touch the screen to stop it. **Tour** is a guided walk through each part, one step at a time (11 steps).
5	The i button	Opens the About box: what the project is, its limits, credits and the creator.
6	Bought and Sold	What you hold with profit or loss on each, and your last six sales.
7	Where your money is	A ring showing how your money splits between stocks, ETFs, bonds, futures, options and cash. With more than two kinds, the key under it scrolls.
8	Leaderboard	Everyone ranked by **percent return**, so a small portfolio can beat a big one. Shows the top four and you.
9	Combined strategy reading	A dial that combines the five strategies' votes (pages 14 to 15).
10	Search stock	Type a company name, even with a small spelling slip, and press Enter. Both NSE and BSE listings work.

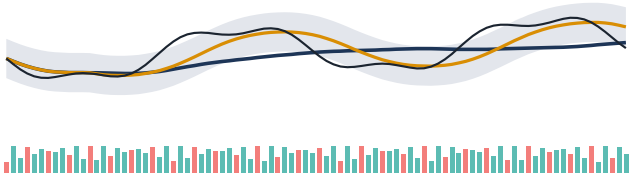
#	What it is	What it does, in plain words
11	Chart controls	1, 2 or 4 charts; Day, Week or Month candles; 6M, 1Y, 2Y or 5Y of history; Fibonacci lines; and a Dark switch for just the chart.
12	The chart	The price history with its tools (next page).
13	Buy, Sell, Exit all	Buy opens the order pop-up. Sell works once you hold something. Exit all sells everything at once; your starting money is not reset.
14	Indicators	Six gauges that read the chart (pages 12 to 14). Hover over the small i beside a name for a description.
15	The strategies table	Five CMT strategies and an ML model, each with its vote and its back-test. A Full view button opens a chart and a plain explanation.
16	The round chat button	Stays in the corner while you scroll. It opens 'Ask me anything', a helper that explains words and how the app works. The small blue document button above it opens this guide.

PART 2. THE SCREEN

The chart and its tools

What the lines, bands and bars on the chart mean. You met candles and volume on pages 4 to 6.

A chart with its tools



- The price (each point is one day)
- 50-day average, the recent mood
- 200-day average, the long-term mood
- Grey band: the price's usual range
- Volume bars: green = price rose that day, red = it fell

A made-up example to show the parts. A real chart has candles instead of one line.

Moving average (50-day and 200-day)

IN SIMPLE WORDS

A day's price jumps about. The **average of the last 50 days** smooths out the bumps, like finding your average marks over a term instead of one test. The 50-day line shows the recent mood and the 200-day line the long-term mood.

IN FINANCE WORDS

A **simple moving average (SMA)** is the mean of the last N closes. It filters noise and shows direction. Price above both lines, with the 50-day above the 200-day, is a classic uptrend. A **golden cross** is the 50-day rising above the 200-day; a **death cross** is the reverse.

ON THE SCREEN

The orange line is the 50-day average and the navy line the 200-day.

The grey band (Bollinger Bands)

IN SIMPLE WORDS

It is the price's **usual lane**, like the lane markings on a road. Most days the price stays inside it. When the price swings wildly the lane gets wider; when it is calm the lane narrows.

IN FINANCE WORDS

Bollinger Bands (John Bollinger) are a 20-day average with bands **2 standard deviations** above and below. They widen with volatility. A close outside the band is unusual for that stock, but in a strong trend it can stay there.

ON THE SCREEN

The grey shaded area around the price line.

Fibonacci levels

IN SIMPLE WORDS

Drop a ball down a staircase. It does not roll straight down: it pauses on certain steps. Fibonacci lines mark the steps where a price often **pauses** after a big rise or fall.

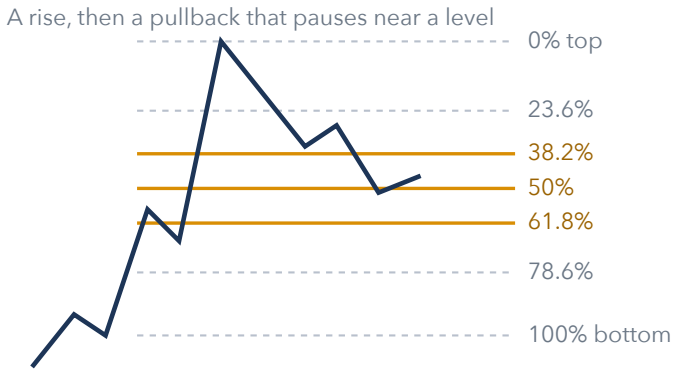
IN FINANCE WORDS

After a swing, prices often **retrace** part of the move before continuing. **Fibonacci retracements** mark 23.6%, 38.2%, 50%, 61.8% and 78.6% of the swing; **extensions** (127.2%, 161.8%) mark possible targets beyond it. They are a guide for support and resistance, not a prediction.

ON THE SCREEN

Switch on **Fibonacci** in the chart controls. The lines are drawn from the highest and lowest price in the period you are viewing.

Fibonacci levels, drawn



Support and resistance

IN SIMPLE WORDS

Support is a floor the price bounced off before. **Resistance** is a ceiling it could not get through. They are like the floor and ceiling of a room.

IN FINANCE WORDS

Support is a price zone where buying has repeatedly stopped declines; **resistance** is a zone where selling has stopped advances. A break through either is often read as a change in the balance of buyers and sellers.

1, 2 or 4 charts; Day, Week or Month

IN SIMPLE WORDS

Like zooming a map: one chart for the big picture, or several side by side to compare views. Day, Week and Month decide whether each candle is one day, one week or one month.

IN FINANCE WORDS

Looking at **multiple timeframes** is standard analysis: the higher timeframe shows the main trend, the lower one the timing. The 1, 2 and 4 buttons set how many charts you see, each in a timeframe of your choice.

PART 3. THE SIX INDICATORS

Indicators, and how to read a gauge

An **indicator** is a number worked out from price (and sometimes volume) that answers one small question about a chart. The six gauges on the right of the screen each answer a different question.

How to read any gauge on the screen

Every gauge is a coloured bar with a black pointer.

- **Red on the left**: this indicator **leans down**.
- **Green on the right**: it **leans up**.
- **Middle**: neutral.

The badge at the top right says *Leans up*, *Leans down* or *Neutral*, and the word under the bar says what the reading is, such as 'Balanced' or 'Stretched high'.

Example: a pointer near the right end



Example: a pointer in the middle



Some indicators measure a **stretch** (RSI, Z-score, the band). For these the bar is turned round: 'stretched high' sits at the red end, because a price that ran up too fast often cools off.

One idea to repeat to visitors. An indicator is a **clue, not an answer**. Six clues can disagree, and that is normal. Never treat one gauge as a promise.

1 RSI (speed)

IN SIMPLE WORDS

RSI is the **speedometer** of a price. It runs from 0 to 100. Near 100 the price has been racing up very fast and may need a rest. Near 0 it has been falling fast and may bounce. But like a fast car, it can keep going.

IN FINANCE WORDS

Relative Strength Index (J. Welles Wilder): RSI = 100 - 100 / (1 + average gain / average loss), using Wilder's smoothing over **14 days**. Above **70** is conventionally **overbought**, below **30 oversold**. In strong trends it can stay at an extreme for a long time, so it is a hint, not a signal on its own.

ON THE SCREEN

'RSI (speed)'. 'Overbought (70)' is the red left end, 'Oversold (30)' the green right end. The word under it says Balanced, Overbought or Oversold.

2 MACD (momentum)

IN SIMPLE WORDS

Two friends run along a track, one quick and one steady. When the quick one pulls ahead, the pace is **picking up**. When the steady one catches up, the pace is **fading**. MACD compares a quick and a slow average of the price in the same way.

IN FINANCE WORDS

Moving Average Convergence Divergence (Gerald Appel): the MACD line is the **12-day** exponential average minus the **26-day** one. The **signal line** is the 9-day average of the MACD line, and the **histogram** is the gap between them. MACD above its signal line means rising momentum, below means fading.

ON THE SCREEN

'MACD (momentum)'. The words are Rising, Slightly rising, Slightly falling or Falling, with the pointer to the right when momentum is rising.

3 Trend strength (ADX)

IN SIMPLE WORDS

ADX is the **wind speed**, not the wind direction. A weak wind means the price is just drifting about. A strong wind means the price is really heading somewhere. Two helper lines say whether the buyers or the sellers are pushing.

IN FINANCE WORDS

Average Directional Index (J. Welles Wilder): a 14-day smoothed measure of **trend strength**, either way. Below about **20** the market is range-bound; above 20 it is trending. Direction comes from **+DI** (upward pressure) against **-DI** (downward pressure).

ON THE SCREEN

'Trend strength (ADX)'. The pointer shows who is in the lead: right with 'Buyers lead', left with 'Sellers lead'.

PART 3. THE SIX INDICATORS

Indicators, continued

Three more gauges, and a word on how they fit together.

1 Z-score

IN SIMPLE WORDS

A Z-score is how far today's price is from its **usual** price, like your height compared with your class. Close to the class average means normal. Far taller or far shorter than everyone means unusual. A price very far above its usual level often drifts back down.

IN FINANCE WORDS

z = (price - 20-day average) / standard deviation over the same 20 days. It measures the price in units of **standard deviations**. Beyond **+2** or **-2** is unusual (about 1 day in 20 if prices were bell-shaped), which mean-reversion traders watch. In strong trends extremes can persist.

ON THE SCREEN

'Z-score'. The words are Stretched high (above +2), Normal, or Stretched low (below -2). It is turned round: stretched high is the red end.

2 On-Balance Volume (OBV)

IN SIMPLE WORDS

Count the shoppers who come into a shop. On days the price goes **up**, **add** the crowd to a score. On days it goes **down**, **take it away**. If the score keeps climbing, the crowd turns up more on the up days. It tells you whether the crowd is busier when the price rises or when it falls.

IN FINANCE WORDS

On-Balance Volume (Joseph Granville, 1963): a running total that adds a day's volume on an up-close and subtracts it on a down-close. A rising OBV suggests **accumulation** (buyers in control); a falling OBV suggests **distribution**. If price makes a new high but OBV does not, that **divergence** is a warning. The app uses the last 20 days' net flow, from -100% to +100%.

ON THE SCREEN

'On-Balance Volume'. The words are 'Buyers backing it' (pointer right), 'Sellers backing it' (pointer left) or 'Balanced'. It is the one gauge that looks at **volume**, not just price.

3 Bollinger band (where the price sits)

IN SIMPLE WORDS

On pages 10 to 11 the grey band was the price's usual lane. This gauge tells you **where in the lane** the price is right now: near the top edge, near the bottom edge or in the middle.

IN FINANCE WORDS

%B = (price - lower band) / (upper band - lower band). Near the **top** of the band the price is **stretched high**; near the **bottom** it is **stretched low**. Outside the band is an extreme. Band width shows volatility.

ON THE SCREEN

'Bollinger band'. 'Top of range' is the red left end, 'Bottom of range' the green right end. The words say Stretched high, Inside range or Stretched low.

Why do the six gauges often disagree?

They answer **different questions**. RSI asks 'how fast?', MACD 'is the pace picking up?', ADX 'is there a real trend?', the Z-score and the band ask 'how stretched?', and OBV asks 'is the crowd behind it?'.

It is like asking six different doctors six different questions. Their answers do not have to match. The skill is in seeing the whole picture.

In finance words: the indicators fall into families: **momentum** (RSI, MACD), **trend strength** (ADX), **volatility and stretch** (Z-score, Bollinger), and **volume** (OBV). Using one from each family is better than using several from the same family, because they add different information.

All six are **lagging**: they are calculated from past prices, so they describe, they do not foresee.

Hover for help

Next to every indicator name is a small **i**. Hover over it (or the badge) for a one-paragraph description. The round chat button in the corner can also answer 'what is the Z-score?' in simple or adult language.

PART 4. THE STRATEGIES

Strategies, votes and the dial

Indicators are clues. A **strategy** is a set of rules that turns clues into a call. Five of them come from the **CMT** (Chartered Market Technician) curriculum, the professional exam in chart analysis.

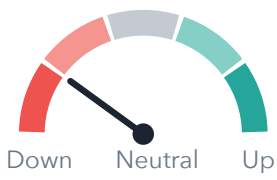
The vote

Each strategy looks at the chart and gives one of three answers, called a **vote**:

- ▲ **Up**: its rules say the reading leans up.
- ▼ **Down**: leans down.
- – **Neutral**: no clear reading. 'n/a' means not enough history to judge.

A vote describes what the rules see today. It is not a promise about tomorrow.

The dial (combined strategy reading)



The dial adds up the five votes (+1 for each Up, -1 for each Down).

It says **Strongly up**, **Leans up**, **Balanced**, **Leans down** or **Strongly down**, and below it: '3 of 5 strategies read down, 1 up, 1 neutral'. The ML model is **not** counted.

Think of five friends voting on whether to take an umbrella. Each looks at the sky in a different way. The dial just counts their votes. It does not know whether it will rain.

1 Fusion stage (the Winner's Circle trend test)

IN SIMPLE WORDS

It asks one question: **is this company in a clear, steady climb and doing better than the market?** Imagine a runner who is both running uphill and overtaking the rest of the class. The strategy likes such runners. If instead the runner is sliding downhill, it votes Down.

IN FINANCE WORDS

From **CMT Level III, Chapter 8** (fusion analysis). A company is in a **clear uptrend** when the price is above a rising 200-day average and the 50-day is above the 200-day, and it also **outperforms the Nifty over six months** (the trend and momentum test). That votes Up. A clear downtrend votes Down. Everything else is Neutral. The **four trend stages** are: clear uptrend, base of an uptrend, base of a downtrend, clear downtrend.

ON THE SCREEN

First row of the table. **Full view** shows a scrollable guide to the four groups, with plain names: **Winner's Circle** (trend, quality and valuation all pass), **Strong contender** (trend plus one of the two), **Momentum only** (trend, but not quality or valuation) and **Watchlist** (the trend has not confirmed).

2 Dow Theory

IN SIMPLE WORDS

Climbing stairs: each step is higher than the last, and each rest stop is higher too. That is an **uptrend**. If the peaks and the dips both keep getting lower, it is like walking down the stairs: a **downtrend**.

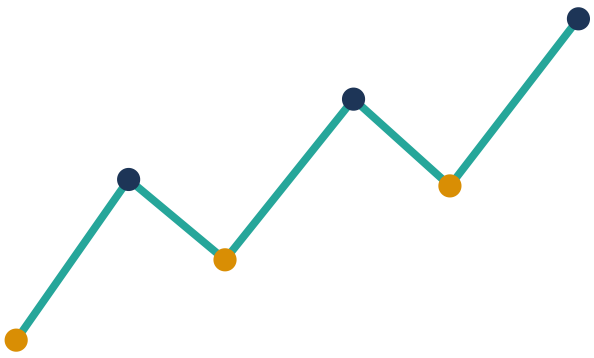
IN FINANCE WORDS

From **CMT Level I** (Charles Dow). An uptrend is a sequence of **higher highs and higher lows**; a downtrend is **lower highs and lower lows**. The trend is **confirmed** only when both agree. The app marks a swing point once five bars on each side confirm it, so it never looks ahead.

ON THE SCREEN

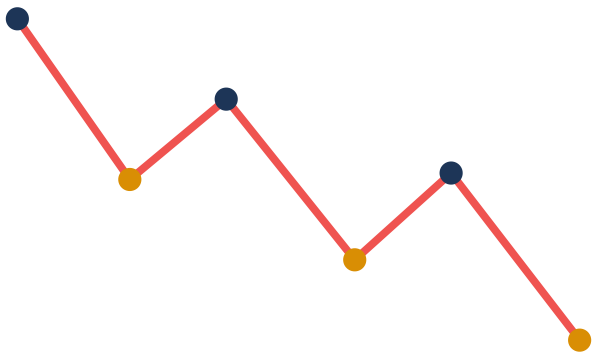
Second row. Votes Up when the last two swing highs and swing lows are both rising, Down when both are falling, Neutral when they disagree.

Higher highs, higher lows



dark dot = a peak, amber dot = a dip

Lower highs, lower lows



dark dot = a peak, amber dot = a dip

PART 4. THE STRATEGIES

Strategies, continued, and the ML model

Three more CMT strategies, and one machine-learning model that is not from the CMT.

1 Relative strength (against the market)

IN SIMPLE WORDS

A race. If the whole class improves by 5 marks and you improve by 10, you did better than the class even though everyone improved. This strategy checks whether a company is **running faster than the Nifty 50**. Leaders tend to stay leaders for a while.

IN FINANCE WORDS

A **CMT core topic**. It divides the stock's price by the index to get a **relative strength line**. It votes **Up** when that line is above its own 50-day average and has risen over three months, **Down** when it is below and has fallen, else Neutral. (Do not confuse it with the RSI indicator, which is different.)

ON THE SCREEN

Third row. The 'What it sees' column says, for example, '-1.9% vs the market (3 months)'.

2 Wyckoff: effort versus result

IN SIMPLE WORDS

Push a heavy door. **Big effort and the door swings open**: a real move. **Big effort and it barely moves**: someone on the other side is pushing back. In markets, effort is volume and result is how far the price moved.

IN FINANCE WORDS

From **CMT Level II** (Richard Wyckoff). **Effort** = volume at least 1.4 times its 20-day average. **Result** = the price move compared with a typical day's range (ATR). Big effort with a big result follows the move (Up or Down). Big effort with a small result is a **stall**: near the low of the recent range that is **absorption** (Up), near the high it is **distribution** (Down).

ON THE SCREEN

Fourth row. 'What it sees' shows, for example, 'Volume 1.8x usual'.

3 Fibonacci confluence

IN SIMPLE WORDS

You saw the staircase on pages 10 to 11. This strategy checks whether the price is **resting on one of the steps** right now. A price that dipped after a climb and is pausing on a step may be getting ready to climb again.

IN FINANCE WORDS

From **CMT Levels I and II**. Over the last six months' swing it checks whether the price sits within 4% of the swing's size from the **38.2%, 50% or 61.8%** retracement. A pullback in an **uptrend** that holds there votes **Up** (support); a bounce in a **downtrend** that stalls there votes **Down** (resistance). 'Confluence' means several clues meeting at one level.

ON THE SCREEN

Fifth row. 'Not at a key level' means Neutral.

4 The ML model (machine learning)

IN SIMPLE WORDS

We showed a computer thousands of days of old prices from 137 companies and let it learn **which clues usually came before a rise or a fall**, a bit like learning which clouds usually bring rain. It then says something like '52% chance of a higher close tomorrow'. But day-to-day moves are mostly luck, so it is only about as good as a **coin toss**.

IN FINANCE WORDS

A **logistic regression** (a simple, explainable model, not deep learning) with **18 inputs**: the five strategy votes plus RSI, Stochastic RSI, Bollinger %B, Z-score, MACD, ADX lean, On-Balance Volume, recent returns, volume, volatility and distance from the 200-day average. It was **trained on the first 70%** of the dates and **tested on the last 30%**, data it had never seen. Out of sample it scored about **50 to 51%**, the same as always guessing the more common direction. It shows a lean only at 54% or more (Up) or 46% or less (Down).

ON THE SCREEN

The sixth row, marked 'ML model'. **Full view** shows a bar chart of which inputs pushed today's number up or down. It is a **second opinion**, not counted in the dial.

Key message for visitors. The strategies and the model are **tools for understanding a chart**, from a real professional curriculum. We tested them honestly and they were about as good as a coin toss at predicting. We show that on purpose. That honesty is the point of the project.

PART 4. THE STRATEGIES

The back-test, and why honesty matters

The last column of the strategies table is the one visitors ask about most.

What is a back-test?

IN SIMPLE WORDS

Before an exam you practise with **last year's question paper** to see how well your method works. A back-test does the same for a strategy: we **replay old prices** and see how often the strategy would have been right. Then we check it again on a part it had never seen.

IN FINANCE WORDS

A **back-test** applies a rule set to **historical data** to measure how it would have performed. A fair back-test avoids **look-ahead bias** (using information that was not available at the time) and **overfitting** (tuning rules until they fit the past). Here the rules were fixed first and the results are reported on the dates held back.

How to read 'Back-test on this company'

IN SIMPLE WORDS

For the company on screen, the column says **how often that strategy's Up or Down vote came true**. '49% next day' means: on the days it said Up or Down for this company, the price moved that way the very next day 49 times out of 100. '41% over 20 days' means the same, about a month later.

IN FINANCE WORDS

Each row's figure is that company's own **hit rate**: the share of Up or Down votes followed by a move in that direction over the next day and the next 20 trading days, measured on that company's own history (roughly 900 to 1,100 calls for Fusion stage, Dow Theory and Relative strength, around 300 for Fibonacci and only about 150 for Wyckoff, which votes on far fewer days). Hover over a number to see how many past calls it rests on and the all-company average for comparison. If a strategy made fewer than 30 calls, the row says 'too few past calls to judge'. For the ML model the figure is from days it was not trained on.

ON THE SCREEN

The header reads 'Back-test on this company: how often it matched the next move'. A company with no saved history shows the all-company average and the header says so.

The honest headline

The next-day numbers are almost all (about 9 in 10) between **45% and 55%**. The 20-day numbers spread wider, but their typical value is still close to 50%. A **coin toss is 50%**.

If the figure is	It means
About 50%	No better than a coin toss
Above 50%	Slightly better, but small samples can fluctuate
Below 50%	Slightly worse than a toss, for this company

Do not read a 53% on one company and a 48% on another as 'one is better'. Per-company numbers wobble by a few points by luck alone.

Why show numbers that are not impressive?

Because it is true. Share prices from one day to the next are very hard to forecast. If a website claimed 80% accuracy you should be suspicious.

This site's promise is different: it teaches how professionals **read** a chart, and it is honest about how far reading goes. That is why the screen says the strategies **describe, they do not recommend**.

So what are the strategies good for?

IN SIMPLE WORDS

A weather forecast cannot tell you if it will rain on your cricket match, but knowing how to read the clouds still helps you decide whether to carry a raincoat. The strategies help you **understand what the chart is doing and why**, and to ask better questions.

IN FINANCE WORDS

In practice technical signals are used as **inputs to a process** that also includes **risk management** (position sizing, stop-losses) and the investor's time horizon, not as stand-alone predictors. The value is in discipline and understanding, not in a guaranteed edge. Past results do not guarantee future results.

ON THE SCREEN

Click **Full view** on any row. A pop-up shows a chart of what that strategy is looking at, a key, and a plain explanation of what it is doing today for this company.

THE PRESENTATION

Using the site, and a 3-minute demo

How a visitor uses it, step by step

1. **Start.** Choose a name (a first name or nickname) and how much practice money. Returning visitors choose 'Returning user' and type the same name. Portfolios have no passwords, so anyone who knows a name can open it.
2. **Search.** Type a company, for example Reliance, and press Enter.
3. **Look.** Read the chart. Try Week, 5Y and Fibonacci.
4. **Read.** Look at the six gauges, the dial and the strategies table. Click one Full view.
5. **Practise.** Press Buy, choose how much, buy. The pop-up stays open for the next order.
6. **Watch.** Your holdings, ring and rank update. Check the leaderboard.
7. **Ask.** Use the round chat button for any word you do not know.

Handy things to know

- **Demo:** press the Demo button and the screen runs itself, with captions and a voice, and at the end it tries the chat helper by itself. The speed buttons on its card (0.75× to 2×) work like a video's playback speed. Use it when nobody is at the screen. Touch the screen to take over.
- **This guide:** the small blue document button above the chat button opens this guide in a new tab.
- **Tour:** the Tour button walks through every part of the screen in 11 steps, each with a fuller explanation. You press Next and Back at your own pace.
- **Dark mode:** good for a dim room or a projector.
- **Dashboard view:** off gives the simple phone-style layout.
- **Leaderboard:** ranked by percent return since the start, so small portfolios can win.
- **Slow or odd?** The site is on free hosting and may take a moment to wake. Prices may be a few minutes behind.
- **An error message?** Stay calm. A grey note appears with a reference number to quote to the admin. It is not your fault.

A 3-minute demo script

Time	Do this	Say this
0:00	Show the whole screen.	"This is a practice stock market. Real prices, pretend money. Nothing here is advice."
0:30	Search a company. Point at the price and the chart.	"Each bar is one day. Green means the price rose, red means it fell. The bars underneath are how many shares were traded."
1:00	Switch on Fibonacci. Point at the orange and navy lines.	"The lines smooth out the bumps. The pale lines are places where a price often pauses."
1:30	Point at the six gauges. Hover over one i.	"Each gauge asks a different question: how fast, how strong, how stretched, is the crowd behind it. Red leans down, green leans up."
2:00	Point at the dial and the table. Click one Full view.	"Five strategies from a professional exam vote Up, Down or Neutral. The dial counts the votes. In our tests they were about as good as a coin toss, and we say so."
2:30	Press Buy and place a small practice order. Close the pop-up.	"Now I practise. My holding and my ranking just updated. Questions? The round button explains any word."

Do

- Say 'practice money' and 'the reading leans'.
- Let visitors try it themselves.
- Say 'I do not know, let us ask the helper' when unsure.

Do not

- Say what a price will do, or give tips.
- Call it investing advice or a money-making tool.
- Hide the 50% honesty. It is your best point.

QUESTIONS AND GLOSSARY

Questions you will be asked

Is this real money?
No. Everything is virtual. Prices are real, the money is pretend.

Can I make money with this?
It is for learning, not earning. It teaches how to read charts. Real trading carries real risk and needs a real adviser.

Should I buy this share?
The site does not tell you that. It shows what the signals read. The decision, and the responsibility, is yours.

Is the prediction accurate?
There is no prediction. The strategies describe the chart, and in our tests they matched the next move about 50% of the time, like a coin toss.

Why does it say 'leans' instead of 'will'?
Because nobody can know the future. 'Leans' means the clues point that way, not that it will happen.

Are the prices live?
They come from Yahoo Finance and can be a few minutes late. When the market is closed it shows the last close.

Why are the futures and options prices 'calculated'?
Real exchange prices for them are not freely available, so the app works them out from the share price with standard formulas. They are learning estimates.

What is the CMT?
The Chartered Market Technician exam, a professional qualification in chart analysis. The strategies here come from its curriculum.

What does the ML model do?
It learned from old prices which clues came before ups and downs, and gives a chance. In tests it was about as good as a coin toss, so it is a second opinion.

Who made it?
Freya Shah, from Mayo College Girls School, using what she has learned, with an AI coding assistant to write the code.

Why is there no password?
Names are not private. Use a first name or nickname and no personal details.

Which indicator is the best?
None on its own. They answer different questions, so the skill is in using them together.

Glossary: every word in one line

ADX. Strength of a trend, either way.

Benchmark. An index used to compare against, like the Nifty 50.

Bond. A loan to a government or company that pays interest.

Bollinger Bands. A band showing a price's usual range.

Call. An option giving the right to buy.

Candlestick. One bar showing open, high, low and close.

CMT. Chartered Market Technician, a chart-analysis qualification.

Diversification. Spreading money to reduce risk.

Dividend. A share of a company's profit paid to shareholders.

ETF. A fund bought like a share that holds a basket of assets.

Exchange. A regulated market where shares trade (NSE, BSE).

Expiry. The date a future or option ends.

Fibonacci. Levels (38.2%, 50%, 61.8%) where prices often pause.

Future. An agreement to buy or sell at a fixed price on a fixed date.

Index. A scoreboard of a basket of shares.

Leverage. Controlling a big position with a small deposit.

Long / Short. Profit from a rise / profit from a fall.

Lot. The fixed quantity in one contract.

MACD. A momentum gauge from two moving averages.

Margin. A deposit that backs a futures position.

Moving average. The average price over the last N days.

OBV. A running total of volume on up days minus down days.

Option. The right, not the obligation, to buy or sell.

Portfolio. Everything you hold, plus cash.

Premium. The price paid for an option.

Put. An option giving the right to sell.

RSI. A 0 to 100 speed gauge of recent moves.

Strike price. The fixed price in an option contract.

Support / Resistance. A price floor / a price ceiling.

Trend. The general direction of a price.

Volatility. How much a price swings.

Volume. How many shares were traded.

Z-score. How far a price is from its 20-day average, in standard deviations.